

A Comparative Evaluation of OLS, Ridge and Lasso in the Presence of Multicollinearity and Outliers

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Abstract

Correlation among the predictors in linear regression often leads to unstable and unreliable estimates. Ridge regression, a biased but effective approach, is widely applied to address this challenge, with numerous methods proposed for selecting the optimal penalty parameter. This study examines 14 prominent ridge parameter estimators, including the top seven from a paper that compared 366 estimators and seven additional methods, benchmarking them against Ordinary Least Squares (OLS) and Lasso. Performance was evaluated under varying degrees of multicollinearity using simulated data, with 25% outliers. Results show that certain ridge estimators perform consistently well with small samples and strong correlations (≈ 0.95) when outliers are present. Moreover, Lasso demonstrated strong performance in larger datasets, except in cases of extreme outliers. Importantly, analysis of real-world data further validated the findings from the simulations, underscoring the practical relevance of the results.

Keywords: Lasso, MSE, Multicollinearity, Outliers, Ridge Regression, Simulation

1. Introduction

Linear regression is a widely used technique for modeling the relationship between a dependent variable and one or more independent variables (Herawati et al., 2024). However, challenges often arise, particularly concerning model assumptions. A key assumption is the absence of multicollinearity; violation of this assumption undermines the reliability of parameter estimation (Gujarati, 2013; Herawati et al., 2018). In multiple regression, multicollinearity occurs when independent variables are highly correlated, leading to inflated standard errors, unstable coefficient estimates, weaker t-tests, and reduced predictive accuracy (Gujarati, 2013).

Addressing multicollinearity is critical, as it can distort inference and increase the risk of incorrect conclusions (Gibbons, 1981). Regularization methods, which shrink coefficient estimates, are effective solutions. Among them, Ridge Regression, Lasso, and Elastic Net (EN) are widely applied (Herawati et al., 2024). Ridge adds bias to stabilize estimates, reducing variance and improving precision relative to Ordinary Least Squares (OLS) (Li et al., 2010; Kibria & Lukman, 2020). By contrast, Lasso and EN tackle multicollinearity by shrinking coefficients of highly correlated predictors toward zero, aiding variable selection and enhancing model robustness (Emmert-Streib & Dehmer, 2019; Liu & Li, 2017).

In 1970, Hoerl and Kennard introduced the concept of ridge regression to address the issue of multicollinearity in engineering data. They discovered that there exists a non-zero value of k (ridge parameter) for which the MSE of the ridge regression estimator is smaller than the variance of the OLS estimator (Hoerl & Kennard, 1970). Over the years, numerous researchers have delved into this area of study, devising and suggesting various estimators for k . To cite a few, Dempster et al., 1977; Gibbons, 1981; Hoerl et al., 1975; Hoerl & Kennard, 1970; J. F. & P, 1976; Khalaf & Shukur, 2005; Kibria, 2003; Kibria et al., 2012a; Månsson et al., 2010; McDonald & Galarneau, 1975; McDonald & Schwing, 1973; Muniz & Kibria, 2009; Walker & Birch, 1988 and very recently Arashi & Valizadeh, 2015a, 2015b; Aslam, 2014a, 2014b; Ayinde & Lukman, 2016; A. V. Dorugade, 2014; Emmert-Streib & Dehmer, 2019; Kibria & Lukman, 2020; Hefnawy & Farag, 2014; Herawati et al., 2018, 2024; Kibria, 2022; Lukman et al., 2018, 2019; Lukman & Ayindez, 2017; Melkumova & Shatskikh, 2017; Mermi et al., 2024; among others.

Mermi et al. (2024) assessed 366 ridge parameter estimators across varying predictors, sample sizes, correlations, and error variances. From these, we focus on seven leading estimators (excluding the robust version) and add seven proposed by Kibria et al. (2003, 2023) for asymmetric populations. This study compares 14 ridge estimators with OLS and Lasso regression, evaluating performance not only by mean squared error (MSE) but also under multicollinearity and outlier conditions. The findings highlight the behavior of estimators in challenging settings by the simulation results.

2. Methodology

2.1 Ordinary Least Squares Method

Let Y be a column vector representing the dependent variable; X be a matrix of independent variables, and $\hat{\beta}_{OLS}$ is a parameter vector. The ordinary least squares estimator of β is obtained as,

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y$$

2.2 Ridge Regression Method

Hoerl and Kennard (1970) first introduced the ridge regression estimator as a solution to the multicollinearity issue. This approach minimizes the sum of squared residuals while applying a constraint on the total sum of the coefficient squares,

$$\hat{\beta}_{Ridge} = \underset{\beta}{\operatorname{argmin}} (Y - X\beta)^T (Y - X\beta) + k \| \beta \|^2$$

where $\| \beta \|^2 = \sum_{i=1}^p \beta_i^2$. Taking the above equation and finding its derivative with respect to β and setting the outcome to zero, we get

$$\hat{\beta}_{Ridge} = (X^T X + kI)^{-1} X^T Y, k \geq 0$$

where k is known as ridge or shrinkage estimator and need to be estimated from observed data.

2.3 Least Absolute Shrinkage and Selection Operator (Lasso)

Lasso is a regression method that simultaneously performs variable selection and regularization to enhance prediction accuracy and model interpretability (Tibshirani, 1996). By shrinking irrelevant coefficients to zero, it retains only the most important predictors, making it a widely used approach for addressing multicollinearity (Lounici, 2008). The process for estimating Lasso follows a similar procedure as the ridge estimator, but the difference is that the squared ℓ_2 norm ($\| \beta \|^2_2$) in the ridge has been replaced by ℓ_1 norm ($\| \beta \|_1$).The Lasso technique seeks to minimize the sum of the squared residuals while imposing a constraint on the total absolute value of the coefficients, ensuring that it remains below a specified constant.

$$\hat{\beta}_{Lasso} = \underset{\beta}{\operatorname{argmin}} (Y - X\beta)^T (Y - X\beta) + \lambda \| \beta \|_1$$

2.4 Ridge parameters

There are numerous ridge estimators available in the literature for different types of models, mostly mentioned in the introduction, to estimate the ridge parameter k . This study selected the top seven estimators from Mermi et al. (2024), excluding the robust estimator. These selected estimators are k_1 through k_4 and k_{10} through k_{12} (Table 1). Additionally, we utilized seven estimators, which are k_5 through k_9 proposed by Kibria (2023) and k_{13} and k_{14} by Kibria and Lukman (2020), specifically tailored for asymmetric populations.

Table 1: Estimators compared in this study

Estimators	References
$k_1 : \max \left(\frac{1}{q_i} \right)$	(Alkhamisi & Shukur, 2007)
$k_2 : \max \left(\frac{s^2}{\hat{\beta}_i^2} \right)$	(Alkhamisi et al., 2006)
$k_3 : \frac{\hat{\sigma}^2}{\min(\frac{\hat{\sigma}^2}{\hat{a}_i^2} + \frac{1}{\lambda_i})}$	(Dorugade, 2014)
$k_4 : \sqrt{p \sum_{i=1}^p \left(\frac{\max(\lambda_i) \hat{\sigma}^2}{(n-p) \hat{\sigma}^2 + \max(\lambda_i) \hat{a}_i^2} \right)}$	(Lukman & Ayindez, 2017)
$k_5 : \text{median} \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_6 : \max \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_7 : \min \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_8 : \text{geometric. mean} \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_9 : \text{harmonic. mean} \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_{10} : \text{median} \left(\frac{\max(\lambda_i) \hat{\sigma}^2}{(n-p) \hat{\sigma}^2 + \lambda_i \hat{a}_i^2} \right)$	(Lukman & Ayindez, 2017)

Estimators	References
$k_{11}: \max \left(\frac{\max(\lambda_i) \hat{\sigma}^2}{(n-p) \hat{\sigma}^2 + \max(\lambda_i) \hat{\sigma}_i^2} \right)$	(Lukman et al., 2018)
$k_{12}: \frac{1}{p} \sum_{i=1}^p \frac{\max(\lambda_i) \hat{\sigma}^2}{(n-p) \hat{\sigma}^2 + \lambda_i \hat{\sigma}_i^2}$	(Alkhamisi et al., 2006)
$k_{13}: \hat{\sigma} p^{\left(1+\frac{p}{n}\right)}$	(Kibria & Lukman, 2020)
$k_{14}: \hat{\sigma} \times \max \left(p^{\left(1+\frac{p}{n}\right)}, p^{\left(1+\frac{1}{p}\right)} \right)$	(Kibria & Lukman, 2020)

2.5 Simulation Design

Since a theoretical comparison was not feasible, we conducted a Monte Carlo simulation in R. The explanatory variables were generated using the following method (Kibria, 2003):

$$x_{li} = \sqrt{1 - \rho^2} u_{li} + \rho u_{l,p+1}, l = 1, \dots, n; i = 1, \dots, p$$

where ρ is the correlation between explanatory variables and $u_{li} \sim N(0,1)$. The dependent variable was generated as

$$Y_l = \beta_0 + \beta_1 x_{l1} + \dots + \beta_p x_{lp} + \varepsilon_l, \varepsilon_l \sim N(0, \sigma^2)$$

We added 25% in the response variable. We considered $p = 10; \rho = 0.9, 0.95 ; n = 20, 30, 50, 100$; and $\sigma = 1, 5$. Each design was replicated $N = 5000$ times. The true coefficients were set to $\beta = I_p$. Model performance was assessed using the averaged mean squared error (AMSE):

$$AMSE(\hat{\beta}^*) = \frac{1}{N} \sum_{i=1}^N (\hat{\beta}_i - \beta)' (\hat{\beta}_i - \beta)$$

where $\hat{\beta}^*$ denotes the OLS, Ridge, Lasso estimator.

3. Simulation Results and Discussion

This section describes the simulation results under different scenarios in the presence of outliers. Table 2 presents the MSE values of the estimators when the correlation among predictors is 0.90 and the data contain 25% outliers. A clear trend emerges: as error variance increases, the MSE also rises, even though the number of predictors is fixed. Under small-sample, high-variance settings, estimators such as k_7 , k_{13} , and k_{14} consistently achieve lower MSE compared to other estimators, highlighting their robustness. In contrast, when the sample size becomes large, the relative advantage of these specialized estimators diminishes, and conventional approaches such as Lasso become attractive options. These traditional methods are easier to implement, require fewer tuning considerations, and still achieve satisfactory performance in large-sample scenarios, making them practical choices in applied contexts.

Table 2: MSE values of the estimators when correlation = 0.90 and 25% outlier

Estimator	P = 10, σ =1				P = 10, σ =5			
	20	30	50	100	20	30	50	100
OLS	7.119	3.515	1.818	1.057	8.134	3.881	2.022	1.117
k ₁	0.953	0.931	0.880	0.886	0.954	0.932	0.925	0.963
k ₂	1.082	0.983	0.896	0.861	0.968	0.949	0.949	0.979
k ₃	6.547	3.315	1.753	1.040	6.445	3.395	1.893	1.092
k ₄	1.447	1.170	0.970	0.806	1.389	1.194	1.004	0.829
k ₅	1.588	1.148	0.920	0.772	1.493	1.115	0.908	0.782
k ₆	3.913	2.224	1.297	0.874	3.641	2.054	1.235	0.844
k ₇	0.985	0.954	0.916	0.889	0.981	0.959	0.946	0.947
k ₈	1.522	1.142	0.919	0.771	1.402	1.085	0.901	0.796
k ₉	2.158	1.412	1.009	0.787	1.971	1.312	0.965	0.780
k ₁₀	3.896	2.705	1.686	1.045	4.065	2.980	1.868	1.103
k ₁₁	3.692	2.679	1.684	1.045	3.998	2.973	1.868	1.103
k ₁₂	4.016	2.756	1.692	1.045	4.201	3.008	1.870	1.104
K ₁₃	0.963	0.935	0.869	0.795	0.952	0.932	0.894	0.854
K ₁₄	0.963	0.935	0.869	0.795	0.952	0.932	0.894	0.854
Lasso	1.875	1.433	1.191	0.932	2.005	1.514	1.229	0.959

The findings in Table 3, where the correlation among predictors increases to 0.95 with 25% outliers, reinforce and extend these observations. For small samples with fewer predictors and lower error variance, estimators k_7 , k_{13} , and k_{14} stand out as the most effective, yielding consistently lower MSE. Once again, the increase in variance leads to a rise in MSE across all estimators; nevertheless, k_7 , k_{13} , and k_{14} remain persistent in small-sample, high-variance situations, while the group, k_7 , k_{13} , and k_{14} , perform well under broader high-variance settings. For large samples, however, the traditional penalization approaches (Lasso) is again competitive and remain appealing due to their established usage and interpretability, even in the presence of outliers.

Table 3: MSE values of the estimators when correlation = 0.95 and 25% outlier

Estimator	P = 10, $\sigma^2 = 1$				P = 10, $\sigma^2 = 5$			
	20	30	50	100	20	30	50	100
OLS	13.045	6.190	2.821	1.337	15.111	6.664	3.113	1.385
k ₁	0.985	0.966	0.911	0.890	0.979	0.957	0.934	0.958
k ₂	1.146	1.032	0.932	0.872	0.995	0.966	0.954	0.974
k ₃	11.875	5.723	2.651	1.298	11.289	5.347	2.728	1.322
k ₄	1.857	1.347	1.049	0.864	1.751	1.305	1.085	0.875
k ₅	1.803	1.208	0.957	0.813	1.763	1.163	0.952	0.800
k ₆	6.338	3.263	1.616	0.986	5.962	2.835	1.496	0.913
k ₇	1.018	0.975	0.932	0.898	1.010	0.974	0.952	0.944
k ₈	1.743	1.228	0.969	0.809	1.674	1.155	0.950	0.808
k ₉	2.931	1.704	1.114	0.843	2.721	1.504	1.056	0.806
k ₁₀	5.469	3.820	2.360	1.303	5.583	4.006	2.612	1.349
k ₁₁	5.335	3.796	2.355	1.302	5.566	4.003	2.610	1.349
k ₁₂	5.632	3.965	2.385	1.303	5.873	4.091	2.621	1.349
K ₁₃	1.032	0.991	0.906	0.815	0.988	0.961	0.915	0.855
K ₁₄	1.032	0.991	0.906	0.815	0.988	0.961	0.915	0.855
Lasso	2.601	1.819	1.402	1.074	2.877	1.898	1.463	1.093

4. Conclusion

In this study, we conducted a comprehensive assessment of multiple linear regression models under conditions of multicollinearity. Ridge regression was adopted as a biased estimation method to improve the precision of coefficient estimates. From the 366 proposed estimators of the ridge parameter, we concentrated on the top seven, supplemented by seven additional estimators recommended in the literature, and benchmarked their performance against classical OLS and modern regularization methods Lasso regression. To emulate practical data challenges, 25% outliers were introduced, and estimator performance was primarily evaluated using MSE.

Our findings revealed essential insights into the selection of ridge parameter estimators across varying conditions. In the presence of outliers, particularly with small sample sizes and significant error variance, the estimators k₇, k₁₃, and k₁₄ demonstrated superior performance, making them the preferred choices in such contexts. Validation through a real-world dataset further reinforced these simulation results.

As sample sizes increased, the performance landscape shifted, Lasso emerging as consistently strong alternatives. It achieved lower MSEs across diverse levels of multicollinearity, although their effectiveness declined when extreme outliers coincided with high variance. Such scenarios underscored the need for judicious estimator selection when handling data with severe contamination.

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